# MTA
## AUDIO EDITOR



# User Manual

Complete guide to using MTA Audio Editor

**MTA Audio Editor 0.3.0-r210**

Build: 20261015171900

Alessandro De Salvo
braket71@gmail.com

https://github.com/desalvo/mta-audio-editor

EUPL-1.2

Digital Audio Workstation with MTA support

Windows macOS Linux Android iOS

MTA Audio Editor is a true **multitrack Digital Audio Workstation (DAW)** and does not require a project to be constrained to an MTA format. The **Multitrack DAW** project format is intended for unrestricted editing, mixing, effects and mastering; MTA8 and MTA16 remain available as dedicated project formats and export targets.

**MTA** means **Multi Track Audio**: a multitrack container that keeps musical parts separately addressable during playback. MTA Audio Editor can import and edit existing files, but it can also **create new MTA8/MTA16 files** from tracks or stems and export the completed project as **MTA, WAV, MP3, MP3+G or FLAC**; individual tracks can be exported separately as WAV, MP3 or FLAC.

This manual covers the complete MTA Audio Editor workflow: access and platforms, projects, import, waveforms, editing, synchronization, mixing, effects, musical metadata, stem separation, MTA8/MTA16 export, desktop clients and mobile apps. Procedures include screenshots, diagrams and practical examples.

**Contents**


1. Quick start
2. Platforms and operating modes
3. DAW workspace
4. Projects, archives and autosave
5. MTA and audio import
6. Waveforms and cache
7. Transport, buffering, synchronization and Follow
8. Non-destructive editing
9. Tracks, roles and MTA slots
10. Live mixer and VU meters
11. Insert FX, presets and controls
12. Auto Mix and master
13. Lyrics, Chords and Markers
14. Stem separation
15. MTA8/MTA16 and device profiles
16. MTA creation, export and slot mapping
17. Android and iOS/iPadOS
18. Windows and macOS
19. Accounts, TOTP, security and sharing
20. End-to-end examples
21. Quick controls
22. Troubleshooting
23. Glossary
24. Stem count
25. Native updates and Demucs models
26. Automatic mobile Demucs model updates
27. Model-driven stem separation
28. Dynamic Demucs model management
29. AI model authentication
30. Synchronized Lyrics and Chords
31. High-accuracy Lyrics/Chords and karaoke video
32. ChordPro and author-society repertoires
33. Zone metronome and personal tools
34. Audio, VST3 and MIDI
35. AI, usage and subscriptions

Technical appendices

# 1. Quick start

Start by choosing how the project is created: empty, imported from an existing MTA, built from individual audio files, or generated from a stereo mix through stem separation. If the final destination is MTA, also choose MTA8/MTA16 and the target device profile.

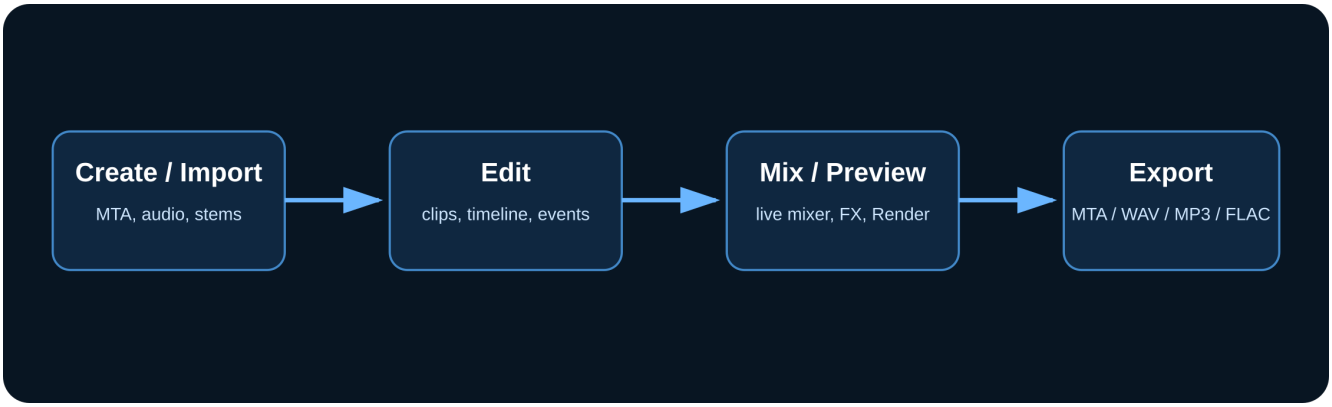



Figure 1 - Project lifecycle. Source audio remains available while clips, events and processing are edited non-destructively.

1. **Open or create.** Use New project, Import MTA or Open project.
2. **Inspect tracks.** Check names, roles, synchronization, channels and waveforms.
3. **Edit and mix.** Work with the timeline, faders, pan, mute/solo and inserts while playing.
4. **Check Render.** Compare dynamic preview with the rendered master.
5. **Export.** Select the device profile and verify Click/Melody and slot mapping.

# 2. Platforms and operating modes

The project model and editor are shared across platforms; file handling differs by client.

| Platform | Scenario | Files | Processing |
|---|---|---|---|
| Browser / server | Centralized and multi-user operation | Server workspace | Server-side FFmpeg, waveforms, stems and renders |
| Docker / Kubernetes | Production and team deployments | Persistent volume / PVC | Server-side |
| Windows / macOS | Local desktop workflow | Native file pickers and bound project archive | Integrated local backend |
| Android / iOS / iPadOS | Mobile access to a server | System file pickers and sharing | Heavy operations remain on the server |

**Mobile first launch:** the apps automatically use their preconfigured service. They do not ask for a server address and never display the built-in default address. Server settings can later store a custom address; only a user-configured custom value may be displayed and edited.

# 3. DAW workspace

The main screen follows a DAW layout: transport at the top, timeline and waveforms in the center, inspector on the right, mixer at the bottom, and musical event/import/export functions around the workspace.

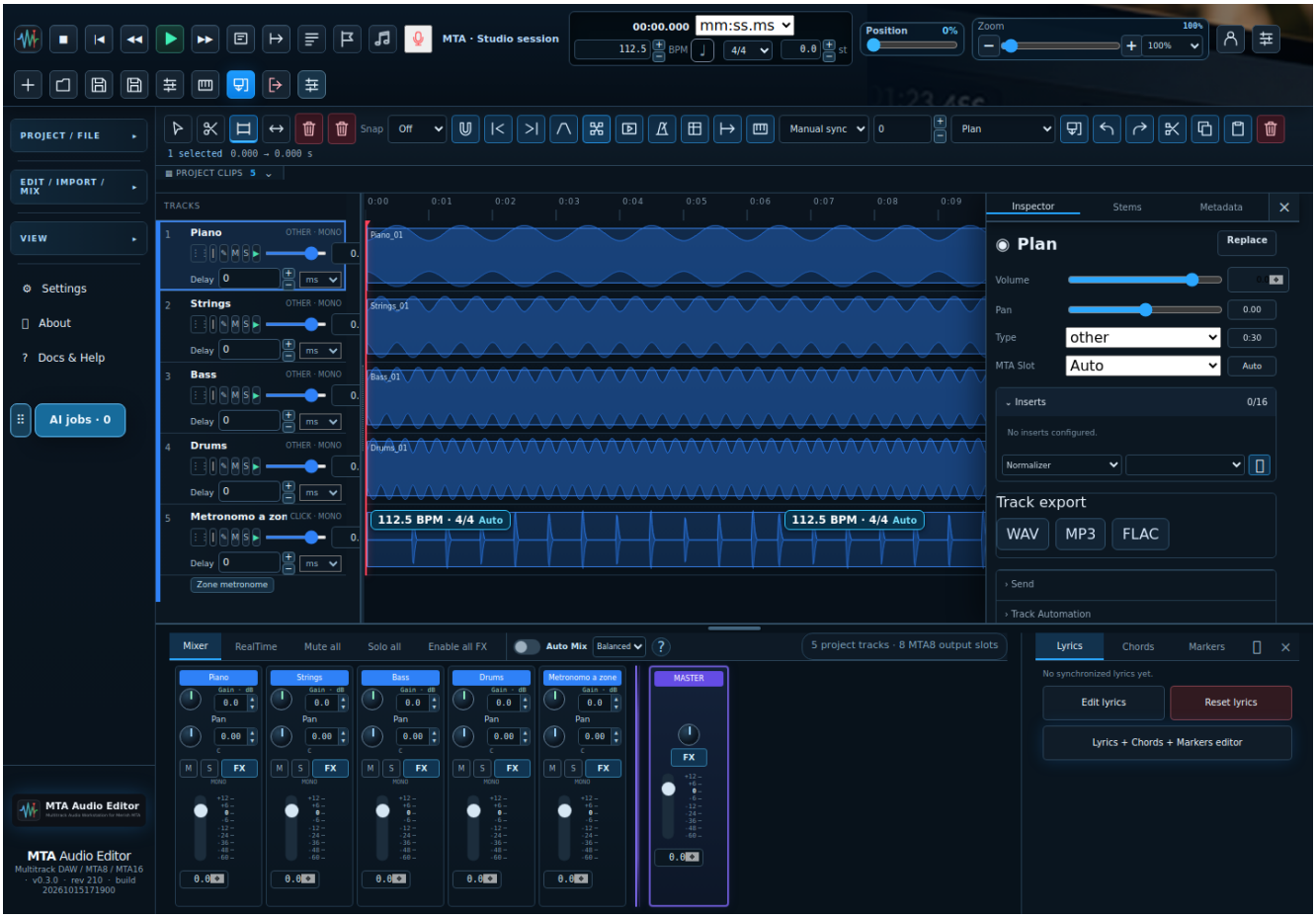
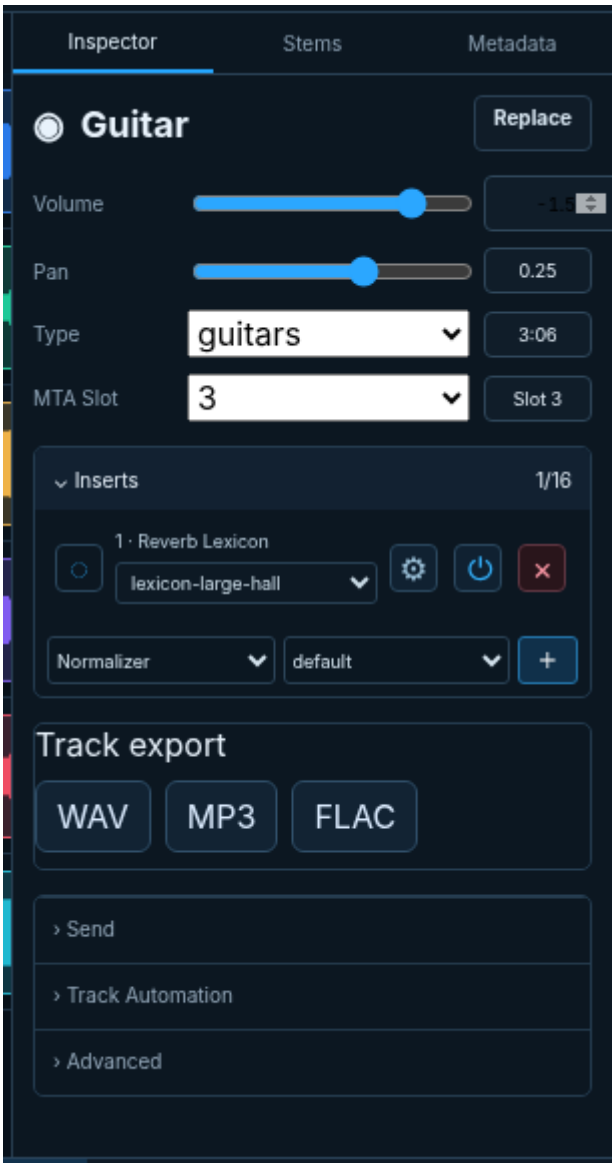



Figure 2 - Real application workstation captured from the current build.




Real application view - Transport: start, rewind, Play/ Pause, tempo, Render, Follow and zoom.




Real application view - Inspector: track properties, level, pan, type, MTA slot and insert chain.

## 3.1 Main panels

The **Timeline** manages clips and time selection. The **Inspector** edits the selected track. The **Mixer** provides a compact view of tracks and master. The larger vertical fader is the volume control; adjacent meters are reserved for realtime signal level, so no redundant static level bar is shown. **Lyrics/Chords/ Markers** manage timed musical events.

## 3.2 Selection and state

The selected track provides context for insert editing, import/replace, split and single-track export. Mute, Solo, level, pan and FX state remain available in the mixer.

# 4. Projects, archives and autosave

The **Multitrack DAW** format is the recommended default for general editing and mixing: it does not constrain the project to 8 or 16 tracks. WAV, MP3, MP3+G and FLAC export directly; when MTA export is selected, MTA8 or MTA16 is chosen and slot mapping is applied only for that output.

A project stores tracks, clips, waveform previews, events, inserts, mix settings, target format, MTA profile, view state and metadata. Source files and required attachments remain in the project workspace.

1. **New desktop project.** Choose a name and save destination; the local archive is bound to the project.
2. **Desktop MTA import.** A project destination is selected before import starts. Cancelling the dialog cancels the import.
3. **Autosave.** Changes are automatically persisted when enabled; manual Save remains available.
4. **Project Info.** Project/File - Project Info shows name, project ID, home/workspace, location, target, duration, track count, artist, BPM, key and MTA profile.

**Project archive versus MTA export:** the project archive contains editing information that does not necessarily belong in the final MTA.

# 5. MTA and audio import

MTA import creates editor tracks from the available audio streams, reads usable names/roles and supported attachments, and keeps the original source. Audio import can add a new track or replace the selected track.

| Operation | Use | Result |
|---|---|---|
| Import MTA | Start from an existing MTA | Project with available streams and metadata |
| Import Audio Track | Add a stem or part | New project track |
| Import & Replace | Replace one part while preserving role/structure | New source on the selected track |
| Import & Separate | Start from a stereo mix | Demucs job and generated stems |

**Example:** select Guitar, choose Import & Replace, check the entry point and Solo it with drums and bass. Fine-tune any offset by moving the track or editing its clip.

# 6. Waveforms and cache

Waveforms are graphical previews, not the audio source. They are calculated when required, stored in the project with a revision and loaded immediately when a project is reopened. Changes that invalidate the preview trigger recalculation.

> **Important:** playback does not depend on waveform availability. Audio can play while waveform generation is still in progress.

# 7. Transport, buffering, synchronization and Follow

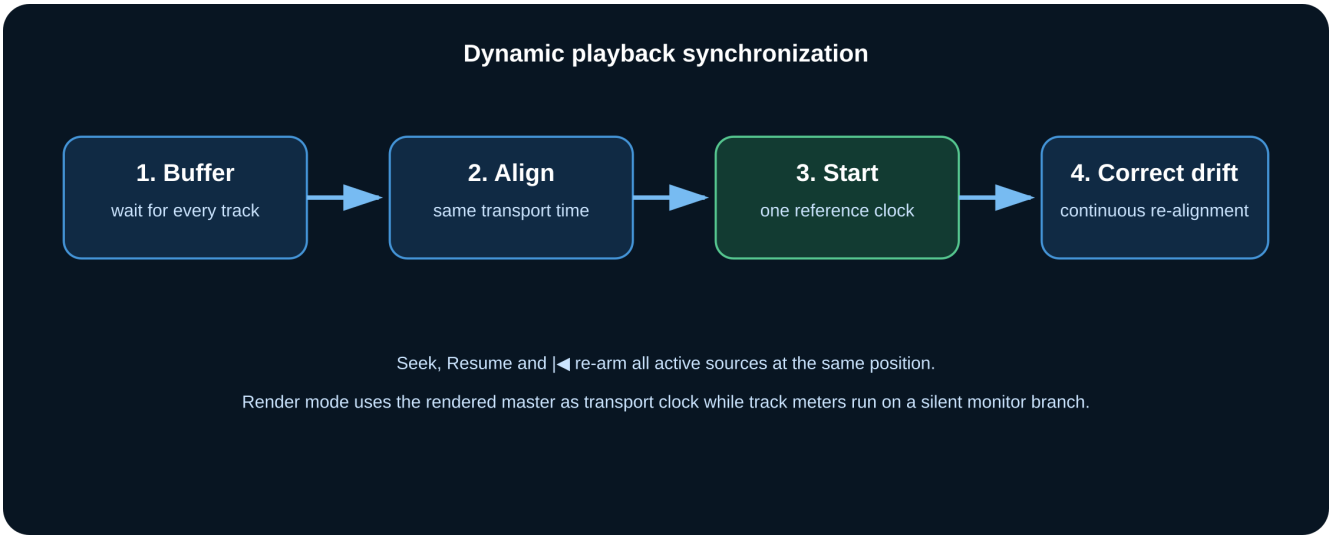



Figure 3 - Dynamic preview startup and synchronization.

The |◀ control returns the transport and active sources to 00:00.000. Rewind/forward change position; Play starts or resumes; Pause keeps the current position; Stop closes the current preview session.

## 7.1 Dynamic preview

Before playback, all tracks are prepared and buffered. The UI reports **Buffering audio... please wait** and then synchronization. Playback starts only when sources are ready. A reference clock monitors drift and realigns active sources when necessary.

## 7.2 Render mode

Render plays a backend-generated master preview. Browser-live parameters remain immediate; changes that are baked into the master trigger an automatic refresh at the current position without a manual Stop/Play cycle.

## 7.3 Follow

Follow keeps the playhead visible in both dynamic and Render modes. In Render mode the rendered master explicitly acts as the transport clock.

# 8. Non-destructive editing

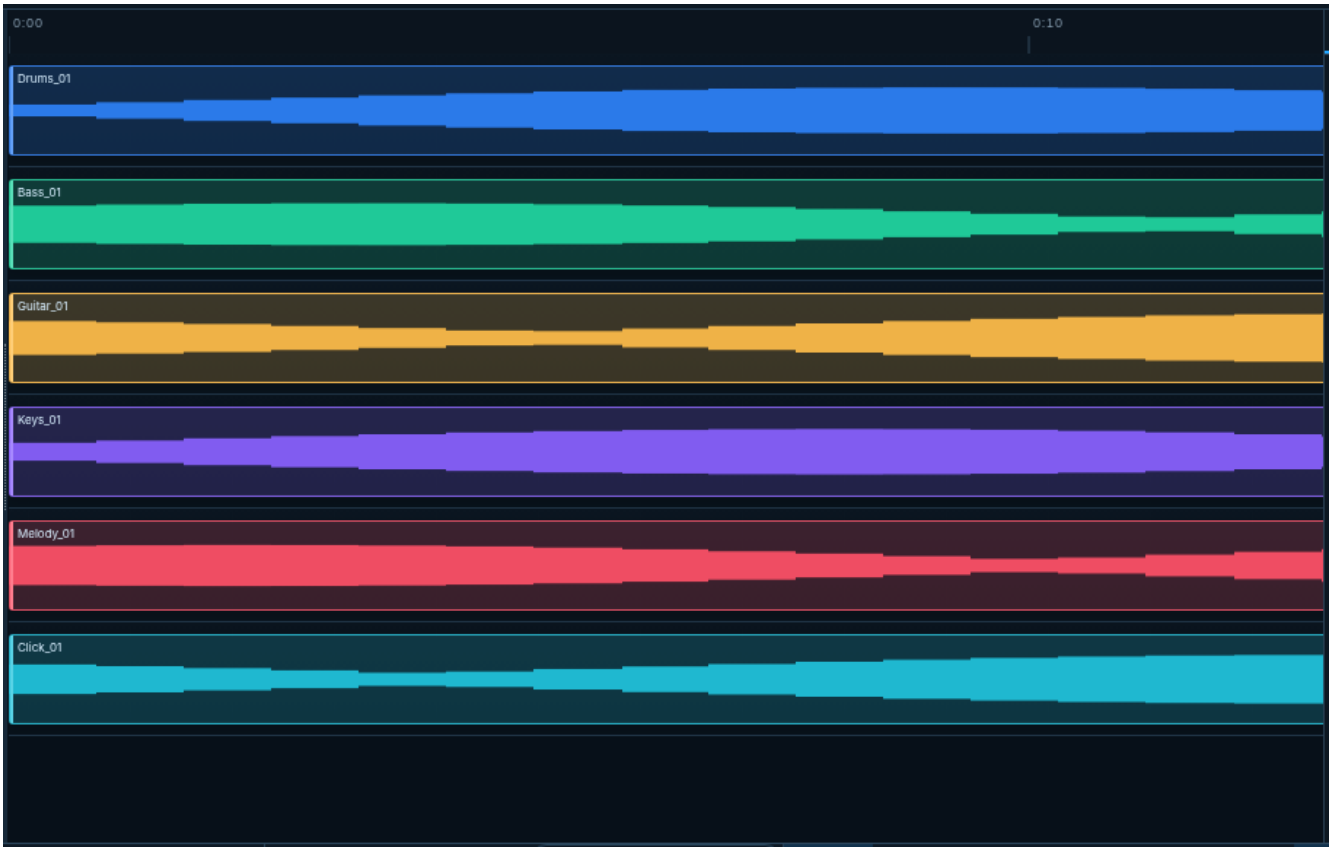



Figure 4 - Timeline and range selection.

Editing changes clip ranges and positions without rewriting the immutable source audio.

| Command | Effect | Typical use |
|---|---|---|
| Split | Split a clip at the chosen point | Create independent sections |
| Remove | Remove a selected interval | Delete noise or a part |
| Ripple | Close the gap by shifting later material | Shorten a section |
| Delete song segment | Global cut plus event realignment | Remove a verse/section |
| Move track | Apply a global time offset | Correct alignment |

# 9. Tracks, roles and MTA slots

Each track contains name, role/type, source, clips, channel information, volume, pan, mute/solo, inserts, waveform state and an **assigned MTA slot**. The Inspector exposes an editable **MTA Slot** field: **Auto** delegates placement to the device profile, while 1...8 or 1...16 stores the preferred output slot. The value is reused as the initial export mapping and is updated when the user confirms a mapping. Multiple tracks may share a slot when they must be mixed into one MTA output stream. Track roles remain important to Auto Mix and mapping; `click` identifies the metronome and `melody` the guide melody.

**Click identification:** explicit metadata and track name/role are preferred. Stream position alone is not sufficient because MTA8 device families use different Click/Melody orders.

### Fixed and adaptive metronome

**Metronome** generates a click at the project BPM. **Adaptive** is intended for songs with accelerando, ritardando or internal tempo changes: every request reanalyses the complete current song from scratch, excludes any existing click track from detection, rebuilds a beat-by-beat tempo map and regenerates the click on the detected beats. A previous adaptive map is never reused.

The track context menu also provides **Metronome from this track** and **Adaptive metronome from this track**. In these modes rhythmic analysis uses only the selected track, which is useful for drums, percussion or other sources with clear transients. Fixed mode estimates BPM and beat phase; adaptive mode rebuilds the complete beat-by-beat tempo map from the selected reference track.

The adaptive metronome can be moved without reanalysis using **Adaptive shift** in milliseconds or beats. The transport can display and accept either absolute `mm:ss.mmm` time or musical `bar:beat` position; conversion uses the fixed BPM/phase grid in fixed mode or the detected beat map in adaptive mode.

## 10. Live mixer and VU meters




Figure 5 - Mixer and meters.

Track level, pan, mute/unmute, solo/unsolo and FX state can be changed while playback continues. Global Mute/Unmute, Solo/Unsolo and FX Bypass/Enable controls simplify mix comparisons.

RealTime meters monitor both tracks and master. In Render mode the per-track analyser path is separated from speaker output, so the master can remain the only audible stream while track meters still receive the real signal.

> **Balancing example:** start playback, establish drums as a reference, balance bass and instruments, use Pan to create space, then check the master. No Stop/Play cycle is required for these changes.

## 11. Insert FX, presets and controls

Tracks and master support chains of up to 16 inserts. Available processors include normalization, compression, limiting, Delay, Lexicon-style Reverb, Room/Ambience, 32-band Graphic EQ, Amplify, Stereo Imager, Maximizer/Loudness, Mastering Wizard, De-Noise and Crackling Cleaner.

| Operation | While playing |
|---|---|
| Bypass / Enable | Applied automatically |
| Preset change | The affected preview is refreshed |
| Custom parameters | Persisted and applied |
| Add / remove insert | The chain updates without manual Stop/Play |

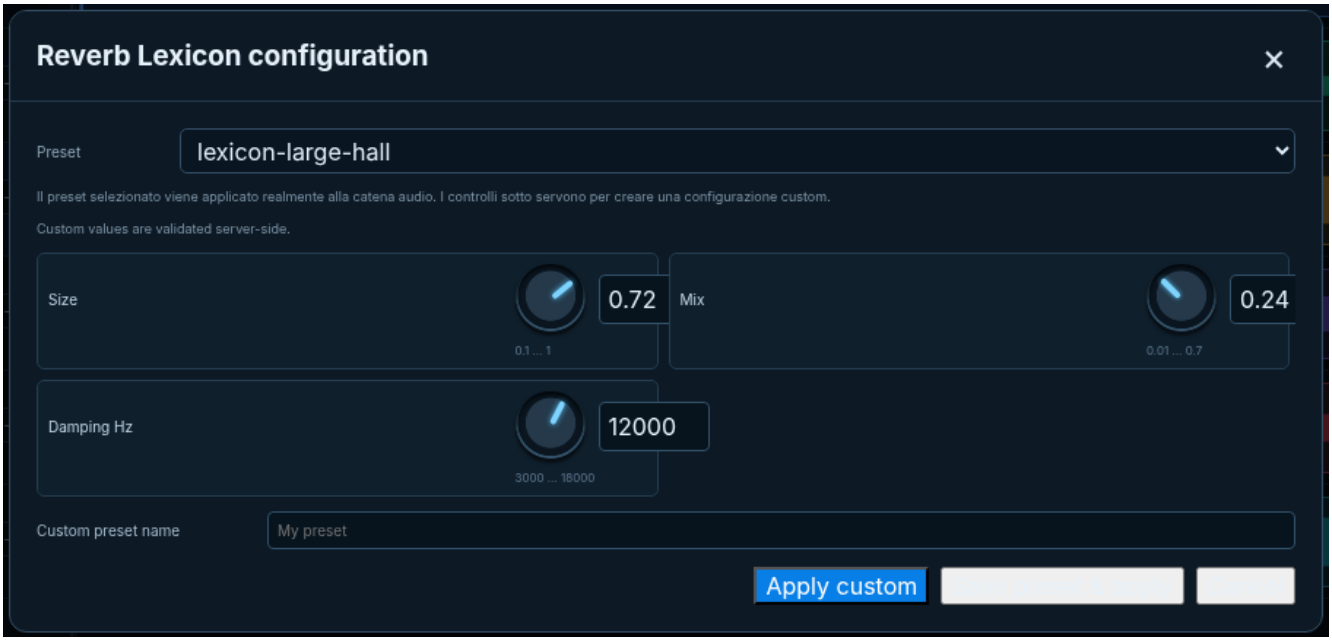



Figure 6 - Real plugin editor: preset selection immediately refreshes the visible parameters and continuous values expose synchronized knob and numeric controls.

Changing a preset updates the plugin's visible parameters **immediately** to values appropriate for that preset: for example Slapback selects a short delay, Large Hall increases reverb size/mix, Wide increases stereo width and Strong raises noise/crackle reduction. Continuous parameters use a synchronized **rotary knob plus numeric input**; moving either control updates the other. A knob commits the value when the adjustment is released; a numeric input commits directly when **Enter** is pressed or when focus leaves the field. The 32-band graphic EQ keeps its dedicated faders. Factory presets can be complemented by user presets, and all parameters are validated by the backend.

# 12. Auto Mix and master

Auto Mix applies deterministic, role-aware balancing rules to track gains, pan and processing. It is reversible: enabling it stores a snapshot and disabling it restores that previous state.

**Recommended workflow:** use Auto Mix as a starting point, then manually refine the result while checking guide vocal/melody, Click, master dynamics and the densest sections of the song.

# 13. Lyrics, Chords and Markers

Lyrics, Chords and Markers are timed project events. Lyrics can be extracted with Whisper large-v3 with word-level timing; chords prefer Chordino/NNLS-Chroma when available and fall back to the internal harmonic analyser. They remain editable, are saved in the project and MTA exports, can be downloaded as text, and can be rendered to PDF with a selectable chord color.

> **Example:** add Intro, Verse, Chorus and Solo markers, timed chord symbols and lyric lines. A global song cut shifts later events together with the audio.

# 14. Stem separation

Import & Separate uses Demucs and can optionally extract synchronized lyrics and chords in the same workflow. For higher accuracy, lyrics extraction uses the isolated Vocals stem when available, while chord analysis uses the original mix. Stems, lyrics and chords are saved into the project and carried into MTA exports.

1. Select the stereo source.
2. Choose Multitrack DAW, MTA8, MTA16 or the current project.
3. Wait for server progress.
4. Rename and classify generated stems.
5. Add Click/Melody if available and synchronize.

# 15. MTA8/MTA16 and device profiles

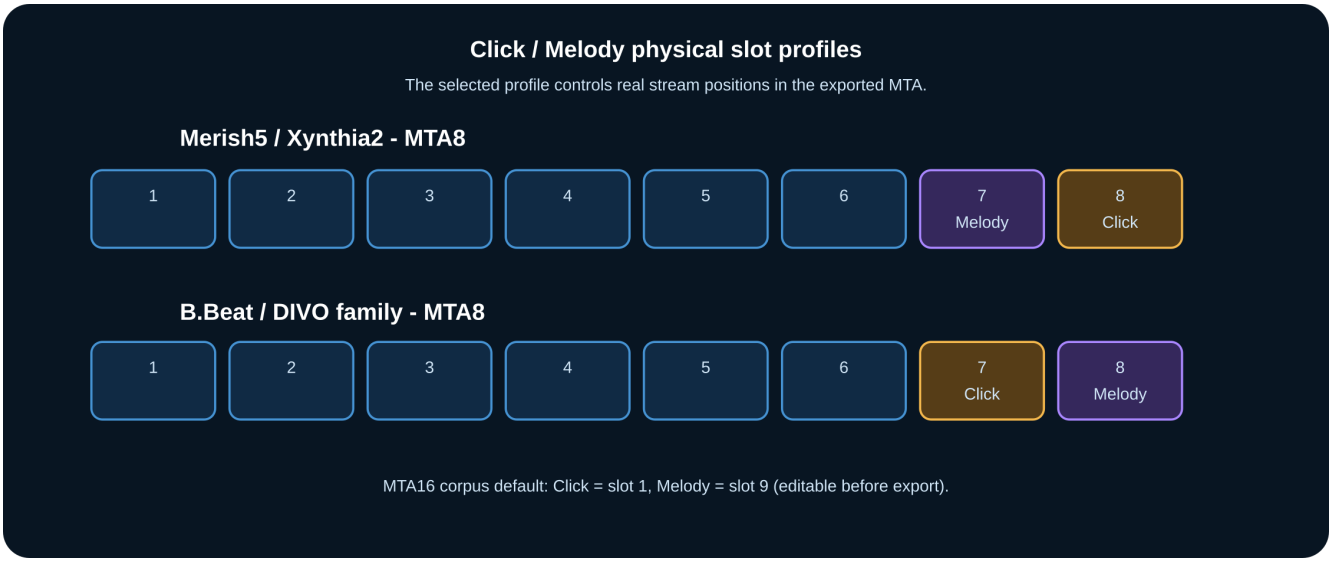



Figure 7 - Physical Click/Melody positions used by export profiles.

**MTA (Multi Track Audio)** is a multitrack format containing audio streams plus metadata/attachments. Device-specific export must preserve the real stream position, not only an internal logical slot label.

| Profile | Click | Melody | Behavior |
|---|---|---|---|
| Merish5 / Xynthia2 - MTA8 | 8 | 7 | Physical positions preserved |
| B.Beat / B.Beat Plus / Evo / Pro16 / DIVO - MTA8 | 7 | 8 | Physical positions preserved |
| M-Live MTA16 - corpus default | 1 | 9 | Suggested and always editable |
| Generic MTA | Manual | | No device-specific assumption |

**MTA16:** the available MTA16 samples consistently place Metronome in stream 1. Explicit Melody Track appears most often in stream 9 and in one sample in stream 10, so slot 9 is a practical default rather than a universal requirement.

# 16. MTA creation, export and slot mapping




Figure 8 - Real Export panel.

The application can **create a new MTA8 or MTA16 from scratch** as well as modify an imported MTA. A completed project can be exported as MTA, WAV, MP3, MP3+G, FLAC or karaoke MP4, while each individual track can be exported separately as WAV, MP3 or FLAC. MP4 export is enabled only when synchronized lyrics exist and can use either a black background or an optional JPG/PNG/WEBP image; word-level timing is used for karaoke highlighting when available. For MTA, select the device profile and verify slot mapping. When a project contains more tracks than the MTA capacity, multiple project tracks can be assigned to one output slot and mixed before container generation.

1. Choose MTA8 or MTA16.
2. Select the target device profile.
3. Check Click and Melody roles.
4. Review or customize slot mapping.
5. Export and test the result on the target device/firmware before live use.

**Silent slots:** when needed, export creates silent intermediate streams to preserve a physical position. Without them, an item assigned to slot 9 could become the fourth real stream.

## 17. Android and iOS/iPadOS

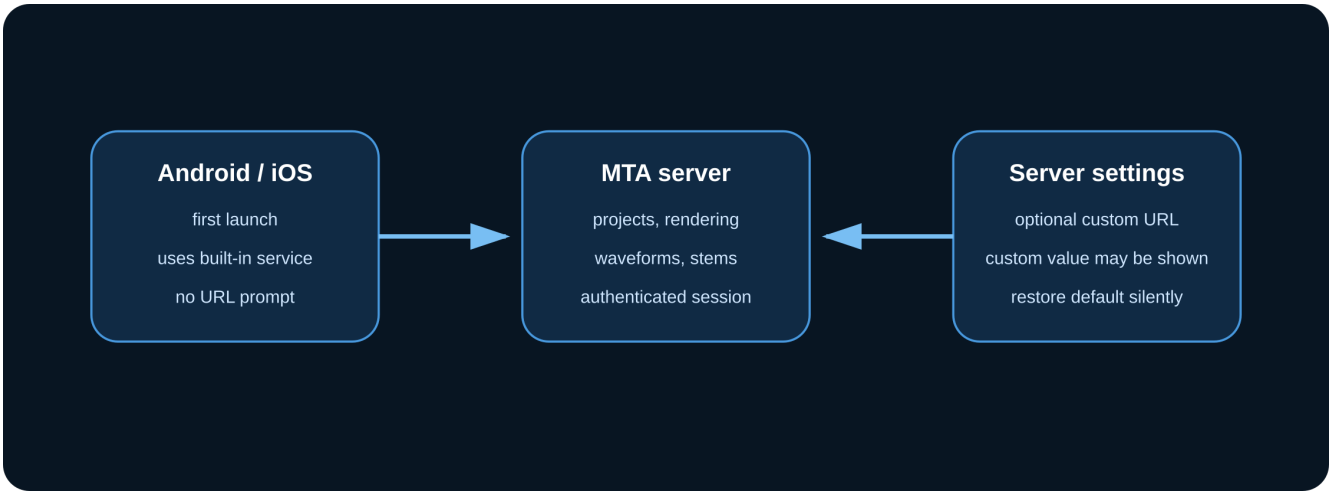



Figure 9 - Mobile connection and optional custom server configuration.

The mobile apps are native shells for the server UI. On first launch they connect automatically to the preconfigured service: installation and first launch never ask for a URL, and the built-in default address is never displayed.

Open **Server settings** only when using a different server. The input can display the custom value chosen by the user; selecting **Use default** removes that custom value and returns to the built-in service without exposing its address.

| Function | Android | iOS/iPadOS |
|---|---|---|
| Import | Storage Access Framework | Document Picker / Files |
| Export | Create Document | Files / Document Picker |
| Sharing | Sharesheet | Share Sheet |
| Session | WebView cookies | WKHTTPCookieStore |

**Offline mode:** the mobile clients do not bundle full FFmpeg/PyTorch/Demucs runtimes. Rendering, stem separation and MTA generation require a reachable server.

## 18. Windows and macOS

Desktop clients run the local backend and native WebView. File choosers are provided by the operating system. macOS packages target both Intel and Apple Silicon; Windows can be packaged with an installer.

A local project archive binding is persisted across restarts. Save As can change its path. Native import/upload limits are configurable in Settings.

# 19. Accounts, TOTP, security and sharing

**Users, passwords, TOTP, email verification, roles and sharing apply to the online/server edition and to mobile apps.** Mobile apps connect to a cloud/remote server because heavy operations - FFmpeg rendering, MTA generation, waveform work and Demucs separation - run server-side; therefore they use the server account. In local native/desktop single-user mode authentication features are unnecessary and hidden, and processing/files stay on the user's computer.

**Recommendation:** public servers should use HTTPS, strong passwords, TOTP where appropriate, regular persistent-volume backups and upgrades validated through the production gate.

# 20. End-to-end examples

## 20.1 Prepare an MTA for a show

1. Import the MTA and save the project.
2. Check tracks, roles, Click/Melody and the device profile.
3. Start Play, wait for buffering, and adjust mixer/pan/FX live.
4. Enable Render and compare the master while keeping Follow and meters active.
5. Review Lyrics, Chords and Markers.
6. Export the MTA and validate it on the target device.

## 20.2 Build an MTA from a stereo mix

1. Create a project and run Import & Separate.
2. Classify the generated stems.
3. Import or create Click and Melody.
4. Correct offsets, then mix and process.
5. Choose MTA8/MTA16 and the device profile.
6. Review slot mapping and export.

## 20.3 Correct a track while the song plays

Solo the target track, change pan/level, bypass an insert and compare presets, then restore Solo. These operations are applied without a manual Stop/Play cycle; Render mode refreshes the required preview automatically.

## 21. Quick controls

| Action | Control |
|---|---|
| Return to start | \|◀ transport button |
| Play / Pause | Transport Play/Pause |
| Undo / Redo | Standard platform shortcuts |
| Global Mute/Solo | All controls in mixer |
| Global FX bypass/enable | Global FX controls |
| Project information | Project/File - Project Info |

## 22. Troubleshooting

| Issue | Check / action |
|---|---|
| Play does not start immediately | Inspect the reported preparation stage and source format; cached compatible audio should start promptly. For C++ inspect underruns and network buffering. |
| Tracks appear out of sync | Seek, Resume or use \|◀; active sources are also continuously drift-corrected. |
| Follow does not scroll | Ensure Follow is enabled. It is supported in Render mode. |
| Meters inactive in Render | Enable RealTime meters; rendered metering is supported. |
| Click routed to the wrong device output | Check track role, MTA profile and physical slot mapping. |
| Waveform missing | Wait for waveform generation; playback remains available. |
| Mobile app cannot connect | If a custom server was configured, review it in Server settings or return to the default service. |
| MTA rejected by hardware | Check MTA8/MTA16 target, profile, slot capacity and target firmware validation. |

## 23. Glossary

| Term | Meaning |
|---|---|
| Clip | A non-destructive time range referencing source audio. |

| | |
|---|---|
| Track role | Musical function used by Auto Mix and MTA mapping. |
| Click | Metronome track. |
| Melody | Guide melody track. |
| Dynamic preview | Current: client track buffers. C++ server: PCM stream with adaptive buffering. |
| Render preview | Backend-generated master preview. |
| MTA slot | Logical and, for profiles, physical stream position in the MTA. |
| Stem | Separated component of a mix. |
| Attachment | Embedded MTA data such as metadata or timed sections. |

## 24. Stem count

Stem separation offers **Auto / 2 / 4 / 6 / 8**. Auto is the default. The current iPhone/iPad build performs Demucs on the server; 8 stems requires a compatible extended backend model.

**On-device Demucs on iPhone/iPad.** Auto mode uses Core ML on the device when a compatible local model is available and falls back to the server otherwise. Models can be downloaded once from the server and reused offline. The current conservative on-device limit is 12 minutes per source file; longer material is processed by the server in Auto mode.

## 25. Native updates and Demucs models

Native Windows, macOS, Android, iPhone and iPad applications check GitHub for newer versions. The **Stable** channel uses tagged releases only, while **Early release** also includes the rolling `early-main` prerelease produced by the latest successful main build.

On iPhone/iPad the app automatically provisions a default 4-stem Demucs Core ML model: it prefers a bundled validated model and otherwise downloads it from the configured server. SHA-256 fingerprints are compared with the server inventory and installed models are refreshed when the fingerprint changes. Once installed, the local model remains available offline.

On iOS/iPadOS application updates must still use TestFlight/App Store or another operating-system-authorized distribution mechanism.

## 26. Automatic mobile Demucs model updates

The Docker/Kubernetes server periodically refreshes Core ML (iPhone/iPad) and ONNX (Android) model repositories from an HTTPS manifest, verifies SHA-256 and replaces files atomically. Mobile clients check the catalogue every six hours and at startup. **Update on Wi-Fi only** is enabled by default, so periodic model updates download only over Wi-Fi unless the switch is explicitly disabled. The baseline 4-stem model is provisioned at the first available network opportunity.

## 27. Model-driven stem separation

Stem counts are no longer limited to 2/4/6/8. The list is generated from models actually available on the server or installed on the device and may include 10, 12, 16, or more stems. Multitrack DAW projects keep all tracks; MTA8/MTA16 constraints are handled only during export.

## 28. Dynamic Demucs model management

The server periodically refreshes the full configured model catalogue, including official Demucs models available in the runtime and Core ML/ONNX artifacts published by the manifest. Mobile and desktop clients download the selected model on demand when splitting is requested. Local options can delete, re-download, or force-update one or more models. Administrators can blacklist a model: managed server artifacts are deleted and the model disappears from the web/client catalogue until it is removed from the blacklist.

## 29. AI model authentication

Demucs downloads use revocable runtime tokens scoped to `models:read`, separate from administrator privileges. Mobile apps obtain and renew the token automatically after a valid session.

## 30. Synchronized Lyrics and Chords

Use a track's context menu and choose **Extract lyrics** or **Extract chords**. Detected events are remapped to the track's actual project timeline, so offsets, clips and cuts are respected. Lyrics and chords become independent project data and remain available if the source track is later edited.

Lyrics can be downloaded as plain text, synchronized text with chords, or as a **PDF**. The PDF places chords above lyric lines and, when available in Project Info, shows title and author/performer at the top. MTA export embeds a lossless synchronized-event representation so the data stays inside the file and can round-trip back into the project.

> **Lyrics extraction:** uses local/upstream OpenAI Whisper when the optional engine is installed. Chords are analysed locally by the built-in engine.

# 31. High-accuracy Lyrics/Chords and karaoke video

**Import & Separate** can extract lyrics and/or chords in the same workflow. Lyrics use Whisper in high-accuracy mode (large-v3 by default, beam search and word-level timestamps). Chords prefer Chordino/NNLS-Chroma when available and fall back to the built-in harmonic analyser.

The Lyrics + Chords PDF lets the user choose a chord colour. **MP4 Karaoke** export renders the project mix audio with synchronized lyrics; an optional JPG/PNG/WEBP background image can be used and chords can also be displayed above the lyrics. MP4 export is unavailable when synchronized lyrics are missing.

# 32. ChordPro and author-society repertoires

Synchronized lyrics and chords can also be exported as **ChordPro** (.cho): title, artist, writers, key and BPM are emitted as ChordPro directives, while chord symbols are aligned into the lyrics using word-level timestamps when available.

**Project Info** can store original title and writers/composers and provides **Search rights repertoires**. SIAE and Soundreef are selected by default. When an official/licensed structured endpoint is configured, candidate works are shown directly and one or more can be selected; when no documented public API is available, the app links to the official repertoire and allows a verified result to be recorded manually. Selected records (society, original title, writers/performers, publishers and identifiers such as ISWC/IPI/work ID) are stored in the project, embedded in MTA exports and printed at the end of Lyrics + Chords PDFs.

## M-Live / Merish MP3 export

The output formats include **M-Live / Merish MP3**, producing a normal MP3 with synchronized lyrics, chords and markers in ID3v2.3/SYLT metadata. Title, performer and authors are embedded when available in Project Info. The technical profile is documented in `MTA-Audio-Editor-M-Live-Merish-MP3-Format-Specification-EN.pdf`.

## Project sample rate and clip conversion

Every project has a **fixed sample rate** selected at creation time: 44.1 kHz (default), 48 kHz, or 96 kHz. It can be changed from Project Info only while the project has no active tracks. Audio imported at 44.1/48/96 kHz is retained in Originals while the timeline copy is automatically resampled to the project rate when required.

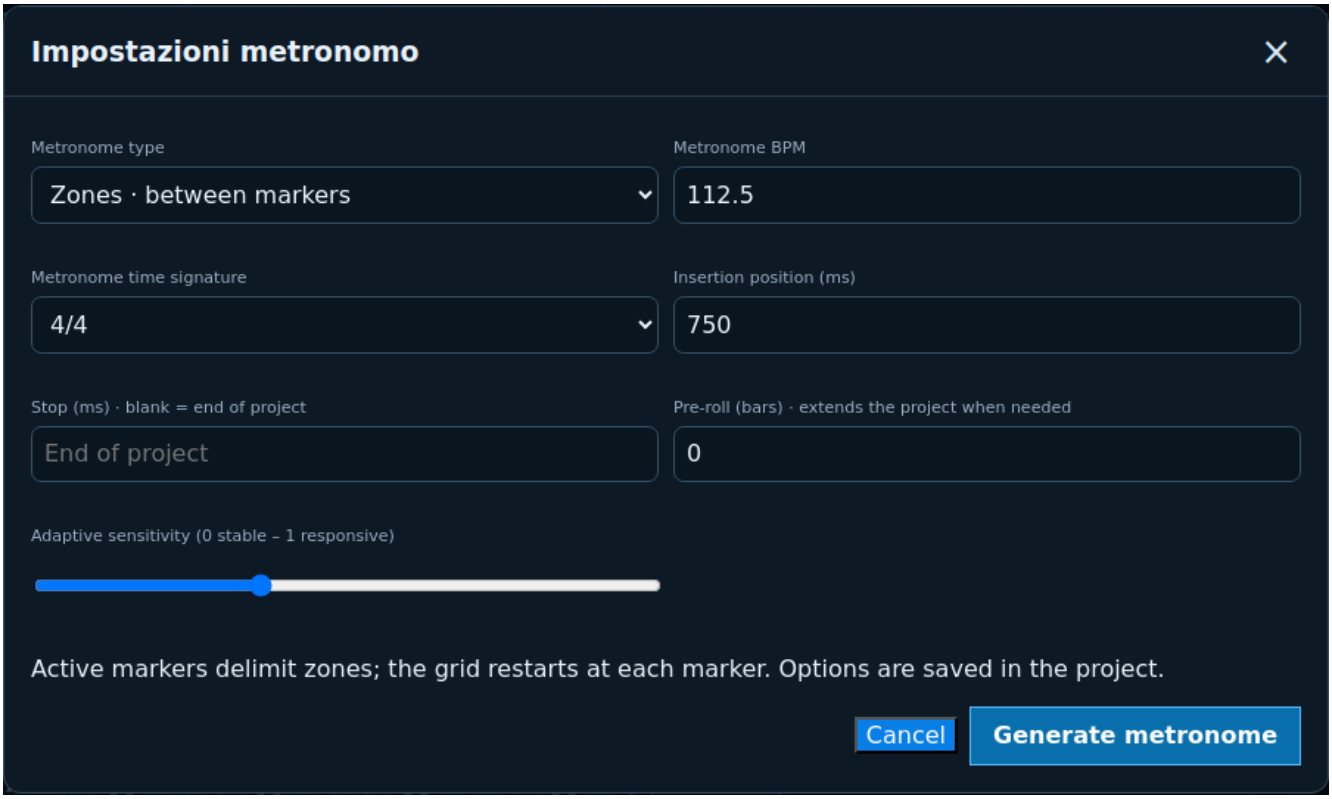
## WAV and MP3+G export

WAV export supports **24-bit PCM** or **32-bit float** at 44.1, 48, or 96 kHz. **MP3+G** export creates a ZIP containing a same-basename MP3 audio file and CDG graphics file with synchronized karaoke lyrics and chords.

# 33. Zone metronome and personal tools

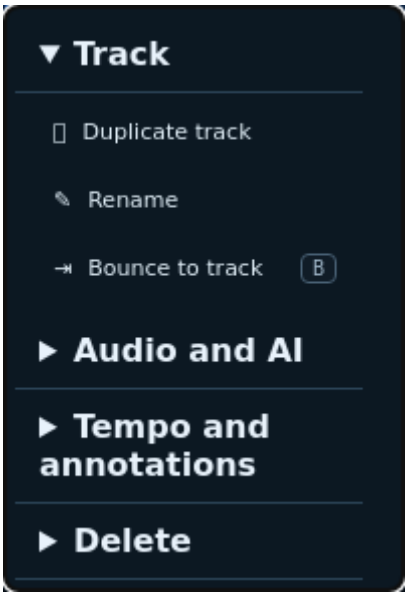
Insert a zone metronome at the current transport position from the context menu, or enter an insertion position in milliseconds in its creation dialog. The first beat matches insertion; subsequent markers restart the grid. Pre-roll adds 0-64 bars before insertion, extending the project at the beginning when

necessary. A Stop limit can mute the end. Moving, adding, disabling or removing a marker regenerates and saves the zone WAV, updating waveform, cache and playback. Manual zone BPM/meter settings follow marker identities.




Metronome creation: mode, independent grid, insertion and pre-roll.

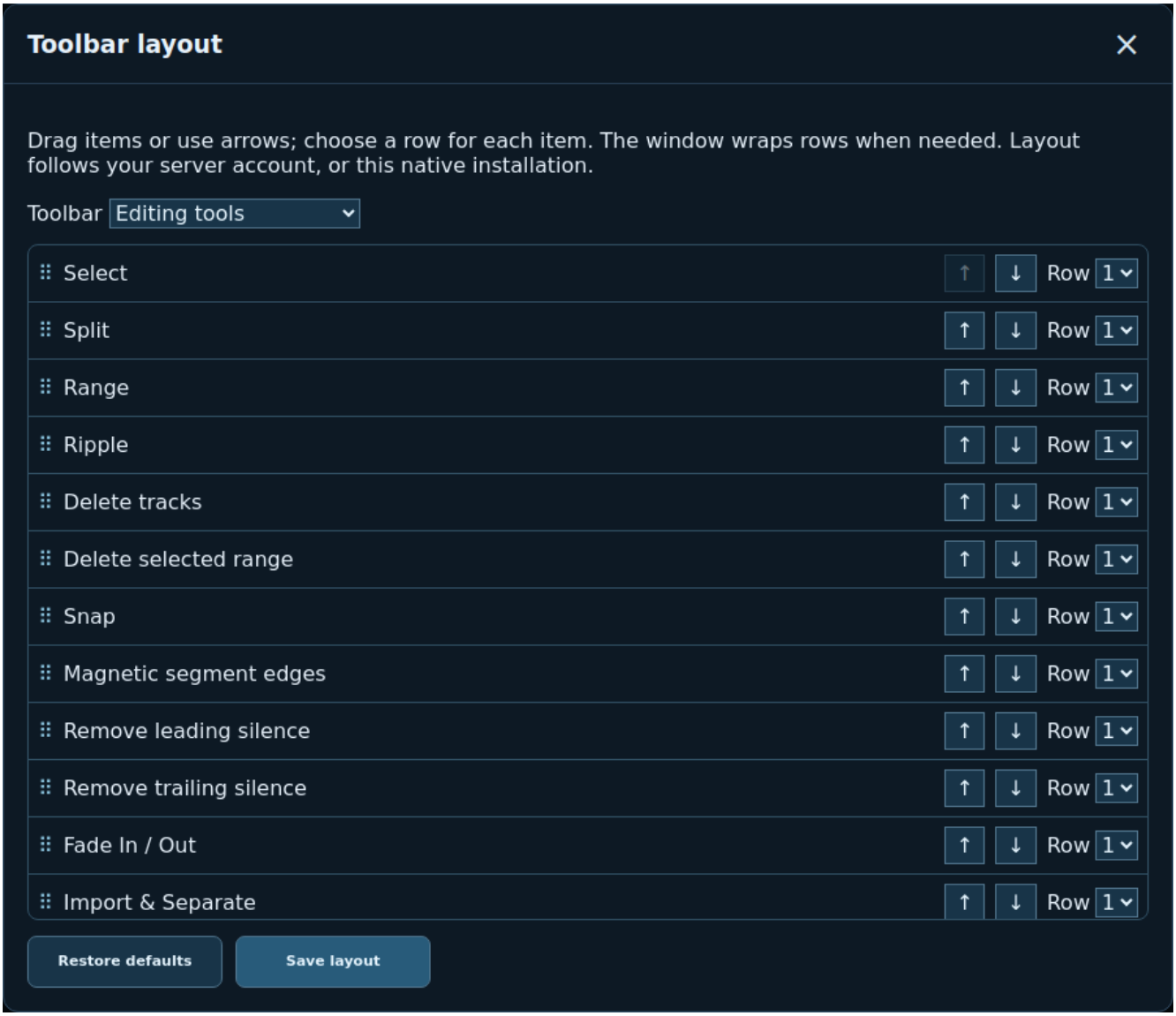
Audio BPM changes project playback speed; Grid changes only metronome BPM/meter, snap and musical display, retaining track speed. Adaptive metronome timing is derived from audio instead. Delay/offset also affects playback; units include milliseconds, beats, bars and bars:beats. R regenerates the selected metronome/chords track; B bounces a track. Snap and magnetic edges guide movement without introducing overlaps on the same track.




Categorized context menu: stays in the viewport and scrolls when necessary.

## Toolbars and AI monitor

Horizontal toolbar buttons show icons; names appear in tooltips and remain accessible to assistive technology. The toolbar configuration button, also available in Settings, opens the item list: choose Transport and project or Editing tools, drag or use arrows to reorder, and assign each item to one of six rows. Save the layout; Restore defaults also resets monitor position. Rows wrap on narrow screens. Timeline zoom retains the time at the view center while dragging and after release; +/− buttons and presets use the same anchor, within project scroll boundaries.




Accessible reordering and row selection.

The AI monitor starts at the left margin, below navigation. Drag its dotted handle to move it; with the handle focused, arrow keys move it and Home resets it. Resizing keeps it in view. Server layout and position belong to the authenticated user; native apps store them in local preferences. Mixer Pan and Gain have dedicated labels; FX opens channel or master effects.




Updated mixer with Gain, Pan and FX.

## 34. Audio, VST3 and MIDI

The Current engine remains selectable. C++ DSP is available in native apps and on the server; server administrators choose it under Web App settings. Browser C++ streaming uses a dedicated PCM worker queue and AudioWorklet; native transport can use the local device. VU meters show measured levels. Complex projects or slow plugins can still cause underruns: inspect diagnostics and reduce load.

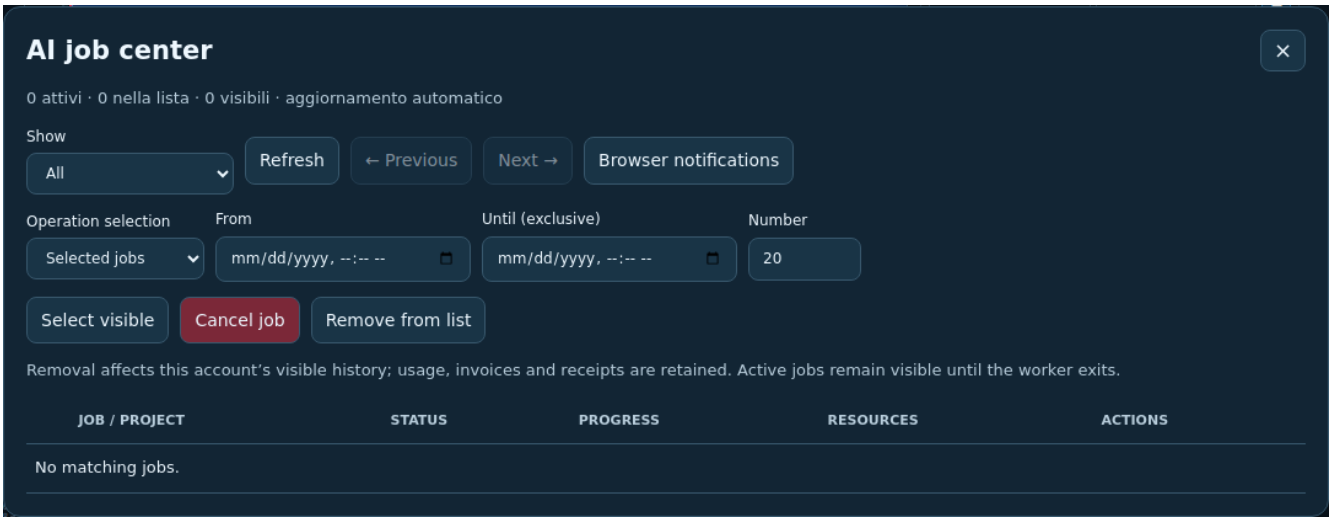
Bounce to track applies selected-track processing; Bounce project to track uses the master mix. VST3 plugins require binaries matching the OS/architecture of the processing computer, including remote use. The optional catalog provides installation, per-plugin progress and controlled uninstall; stored project presets are retained. Close active plugin instances before removing their package. The IT/EN VST3 manual details original editors, VSTi, MIDI and host synchronization.

Online VST3 manual · PDF VST3

## 35. AI, usage and subscriptions

Server AI engines use the global queue. Native apps use one token per remote instance, associated with a user. AI jobs exposes status, CPU/GPU, progress and cancellation even after closing the originating window. Completion notices appear when the origin is closed and results have been saved in the project. Configured SMTP enables job notifications and usage summaries. GPU estimates are identified as estimates; measurement errors must not be interpreted as zero usage.

CPU/GPU consumption accepted by the server is accounted for quotas and billing. Live counters remain separate until job completion and acceptance. Source evidence is retained for audit; missing or invalid data does not authorize charges.



AI job center: status, resources and background controls.

Profile Limits and Quotas collects storage, CPU/GPU, billable usage and history. Subscriptions and credits are available only when enabled by the administrator for that user. Credits cover active-subscription overage, not use without a subscription. An admitted job completes even if its credits run out, without charges beyond its admitted budget. A job requiring overage credits is rejected when they are zero: purchase credits if enabled or contact an administrator. PayPal requires server configuration; it does not imply automatic PayPal renewal.

# Technical appendices

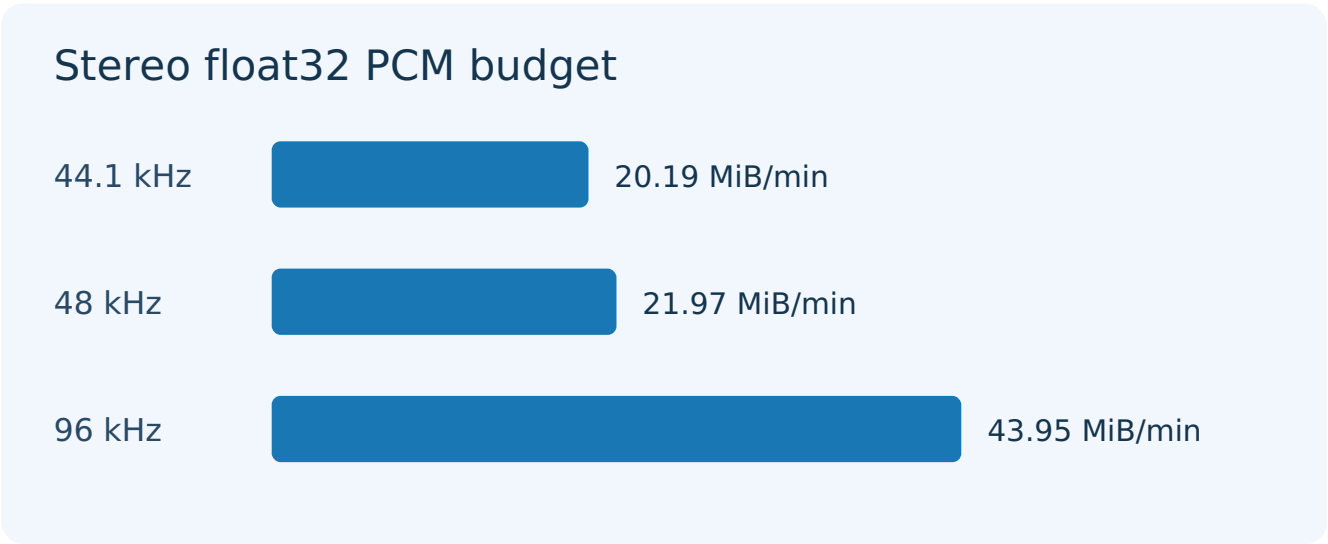
Documented state: 0.3.0-r210. Sizing guidance is an operational estimate, not a certified minimum. Local results do not certify physical devices, plugins or production servers.

## A. Application types and capabilities

| Type | Audio / DSP | AI | Plugin / MIDI |
|---|---|---|---|
| Web + server | Browser Web Audio; server C++ DSP | Global server queue; tokens for remote clients | Server Linux VST3, remote editor; Web MIDI where supported |
| Windows desktop | Current or C++; device transport | Local engines or token-authenticated remote server | Compatible Windows VST3; native MIDI |
| macOS desktop | Intel / Apple Silicon; Current or C++ | Local models or server; accelerator support depends on engine | Matching-architecture macOS plugins; native MIDI |
| Linux desktop | Current or C++; local device | CPU / CUDA / ROCm depending on runtime and hardware | Linux VST3; ALSA/MIDI subject to availability |
| Android / iOS / iPadOS | Mobile shell; browser/ WebView audio | Server; on-device paths require a compatible model/ runtime | Server-side plugins; browser MIDI depends on platform |

## B. Resources and sizing

| Profile | CPU / RAM | Disk / network | Note |
|---|---|---|---|
| Browser | 4-8 GiB device RAM: planning baseline | HTTPS; stable network for streams and remote editors | 256 MiB PCM cache; no guarantee for obsolete WebViews |
| Desktop / DSP | 4 cores / 8 GiB starting point; 16+ GiB for dense sessions | SSD recommended; allow sources, exports and models | Plugins, sample rate and channel count determine actual load |
| Server editor | 4 vCPU / 8 GiB for small tests; size for concurrency | Persistent volume, backups, bandwidth for concurrent sessions | Not a guaranteed user capacity |
| Server / desktop AI | 16-32+ GiB RAM initial budget; model dependent | Separate weights/runtimes: several GiB, sometimes tens | VRAM/unified memory and precision depend on model; validate each engine |
| Mobile | Browser memory/CPU; avoid oversized projects | Stable Wi-Fi recommended for audio and remote plugins | Server DSP does not become phone-local |




Theoretical PCM memory per minute, stereo float32 (excluding graphs, models and duplicate caches).

```
PCM bytes = seconds × sample rate × channels × 4
```

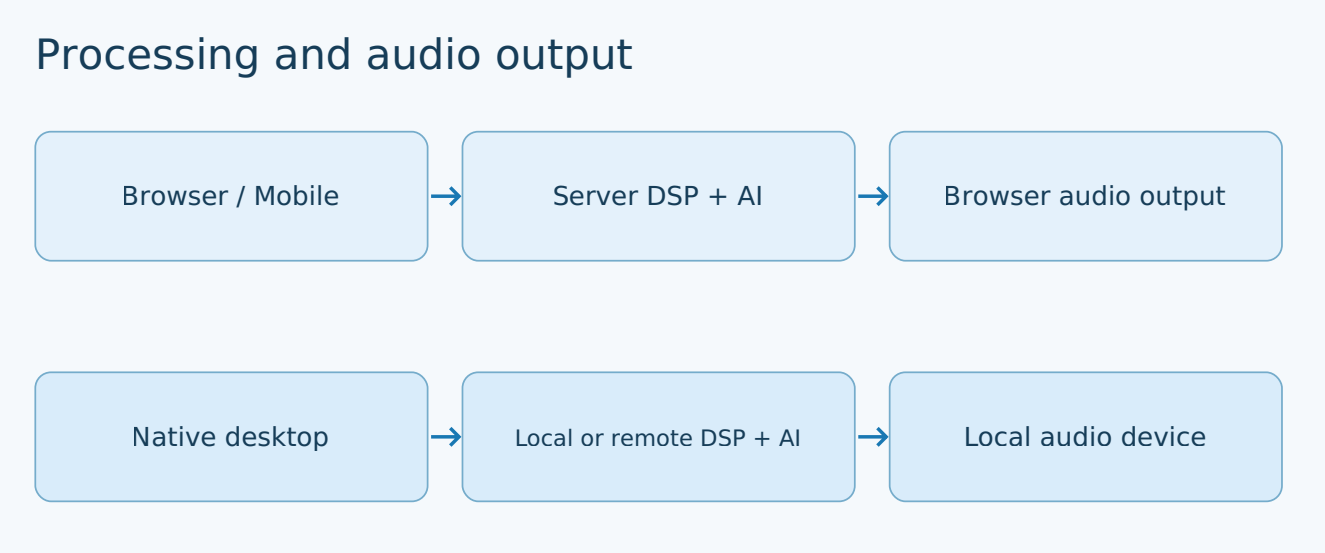
## C. Limits and behavior

| Area | Limit / behavior |
|---|---|
| Project | 44.1 / 48 / 96 kHz; sample rate fixed after first track. Timeline up to 24 h. |
| Metronome | Standard, marker zones, adaptive. Pre-roll 0-64 bars; grid BPM 20-400 including decimals. |
| FX | Up to 16 inserts per chain; validate dependencies and VST3 compatibility per plugin. |
| MIDI | Type 0/1 SMF up to 8 MiB. UMP/MIDI 2.0 depends on driver/ provider, not guaranteed for every device. |
| AI | Global queue; configurable artifact limits and timeouts. An installed model does not guarantee GPU compatibility. |
| Streaming C++ | Adaptive buffer 0.35-1.2 s plus block; not a guarantee of professional low-latency monitoring. |
| Storage / billing | New users default to 5 GiB; admins have no default quota. Quotas/credits depend on plan and permissions. |

## D. Observed performance and validation

Measurements from the preceding local checkpoint: Chromium, eight 30-second stereo tracks; test machine and load do not represent the production server. Original reports remain in the project validation directory.

| Test | Observed result | Local source |
| --- | --- | --- |
| C++ playback | Start 1.239 s; fader response 0.538 s; 0 underruns | r210-continuous-pcm-browser.json |
| C++ jitter / UI stalls | Network delays up to 180 ms, five 450 ms UI stalls: 0 underruns | r210-continuous-pcm-stress.json |
| Current engine | Start 2.198 s; cached restart 0.265 s | r210-continuity-current-engine.json |




Processing and audio output locations.

Certification still depends on real-platform checks: audio interface, MIDI driver, third-party plugins, GPU and proxy. Accepted consumption is accounted; source and validity remain in the audit; zero consumption does not prove no GPU use.